



Homework 1

ggplot2 · dplyr Fundamentals

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Directions

- Submit **one Quarto document** (`.qmd`) to Brightspace:
 - `dan1-310-hw1-LASTNAME-FIRSTNAME.qmd`
(e.g., `dan1-310-hw1-choe-byeonghak.qmd`)
- **Due:** February 18, 2026, 11:59 P.M. (ET)
- For **visualization questions**, you must provide:
 1. the **ggplot2 code**, and
 2. a **written comment** (2–4 sentences) interpreting the corresponding figure.
- Unless a question says otherwise, use **dplyr** verbs (`filter()` , `distinct()` , `select()` , `mutate()` , `group_by()` , `summarise()` , `arrange()` , `count()` , etc.) and `ggplot2` .



Setup

```
library(tidyverse)
library(skimr)
```

Part 1. Data Visualization & Summaries

A. Orange Juice Promotions (oj)



Use the following dataset for Questions 1–6.

```
oj <- read_csv("https://bcdanl.github.io/data/dominick_oj_na.csv")
```

Question 1. Quick inspection

1. Print the first 10 rows of `oj`.
2. Use `skimr::skim()` (or another method) to summarize the variables.

```
# YOUR CODE HERE
```

Question 2. Brand-level descriptive statistics (filter + group summary)

Filter to the three brands: **tropicana**, **minute.maid**, and **dominicks**. Then compute **mean** and **standard deviation** of:

- `sales`
- `price`

for each brand.

```
# YOUR CODE HERE
```

Write 2–3 sentences comparing the three brands based on your results.

Question 3. Remove missing values

Create a new data frame `oj_no_na` that removes observations with **missing** values in **either** `price` or `sales`.

1. Show the number of rows in `oj` and `oj_no_na`.
2. Report how many rows were removed.

```
# YOUR CODE HERE
```

Question 4. Price distribution by brand (ggplot + comment)

Using `oj_no_na`, make a figure that compares the **distribution of price across brand**.

- Choose ONE main approach (e.g., boxplot, violin plot, density plot, histogram with facets).
- Include clear axis labels and a readable title.

(a) Provide your **ggplot2** code:

```
# YOUR CODE HERE
```

(b) **Comment (2–3 sentences):** What differences do you see across brands (center, spread, skewness, outliers, etc.)?

Question 5. Log–log price–sales relationship by brand (ggplot + comment)

Using `oj_no_na`, visualize how the relationship between:

- `log10(sales)` and `log10(price)`

varies by **brand**.

- Use a scatter plot with transparency (e.g., `alpha = 0.3`).
- Add a fitted line (e.g., `geom_smooth(method = "lm", se = FALSE)`), and
- Use faceting OR color to distinguish brands.

(a) Provide your **ggplot2** code:

```
# YOUR CODE HERE
```

(b) **Comment (2–3 sentences):** Do you see evidence that higher prices are associated with lower sales? Does the pattern differ by brand?

Question 6. Log–log relationship by brand and ad status (ggplot + comment)

Now extend Question 5 by incorporating `ad_status`. Visualize how the relationship between:

- `log10(sales)` and `log10(price)`

varies by **brand** and **ad_status**.

- Use faceting (`facet_grid()` or `facet_wrap()`), and/or
- Use color/shape for `ad_status`.

(a) Provide your **ggplot2** code:

```
# YOUR CODE HERE
```

(b) **Comment (2–3 sentences):** How does advertising status appear to shift the relationship (level shift, slope change, or no clear difference)?

B. MLB Batting Trends (`mlb_bat`)



Use the following dataset for Questions 7–8.

```
mlb_bat <- read_csv("https://bcdanl.github.io/data/MLB_batting.csv")
```

Question 7. Create yearly hit percentages (data transformation)

Create a data frame `mlb_hit_pct` that contains **yearly hit percentages** for each `hit_type` (Single, Double, Triple, HomeRun).

Your final table should have:

- `year`
- `hit_type`
- `hit_pct` (a percentage or proportion)


```
# YOUR CODE HERE
```

Question 8. Visualize trends (ggplot + comment)

Make a figure that shows the yearly trends in hit percentages for each `hit_type`.

(a) Provide your `ggplot2` code:

```
# YOUR CODE HERE
```

(b) **Comment (2–3 sentences):** Which hit types are increasing/decreasing over time? Any notable breaks or eras?

C. Titanic Survival (`titanic`)



Use the following dataset for Questions 9–13.

```
titanic <- read_csv("https://bcdanl.github.io/data/titanic_cleaned.csv")
```

Question 9. Two-way count table

Create `titanic_class_survival` that counts passengers by `class` and `survived`.

```
# YOUR CODE HERE
```

Question 10. Age distribution by class and gender (ggplot + comment)

Visualize how the **distribution of age** varies across **class** and **gender**.

(a) Provide your **ggplot2** code:

```
# YOUR CODE HERE
```

(b) **Comment (2–3 sentences)**: What differences do you see across class and gender?

Question 11. Survival rate by class and gender (data + ggplot + comment)

1. Create a summary table with the **survival rate** (proportion survived) by **class** and **gender**.
2. Visualize the survival rates.

(a) Provide your **dplyr** code for the summary table:

```
# YOUR CODE HERE
```

(b) Provide your **ggplot2** code:

```
# YOUR CODE HERE
```

(c) **Comment (2–3 sentences)**: Which groups have the highest/lowest survival rates?

Question 12. Conditional distribution of survived (ggplot + comment)

Create a plot that shows the **distribution of survived** across **class** and **gender** using proportions (not raw counts).

(a) Provide your **ggplot2** code:

```
# YOUR CODE HERE
```

(b) **Comment (2–3 sentences):** How does using proportions (instead of counts) change your interpretation?

Question 13. Short interpretation

Write 3 **bullet-point insights** supported by your tables/figures in Questions 9–12.

D. NYC Dog Licenses (`nyc_dogs`)



Use the following dataset for Questions 14–16.

```
nyc_dogs <- read_csv("https://bcdanl.github.io/data/nyc_dogs_cleaned.csv")
```

Question 14. Breed frequency table

Create `nyc_dogs_breeds` with:

- non-missing `breed`,
- `n >= 2000`, and
- sorted by `n` descending.

```
# YOUR CODE HERE
```

Question 15. Visualize the distribution of breeds (ggplot + comment)

Using `nyc_dogs_breeds`, make a figure that shows the distribution of breed counts.

(a) Provide your `ggplot2` code:

```
# YOUR CODE HERE
```

(b) Comment (2–3 sentences): Is the distribution concentrated in a few breeds or spread out?

Question 16. “Top breeds” focus (ggplot + comment)

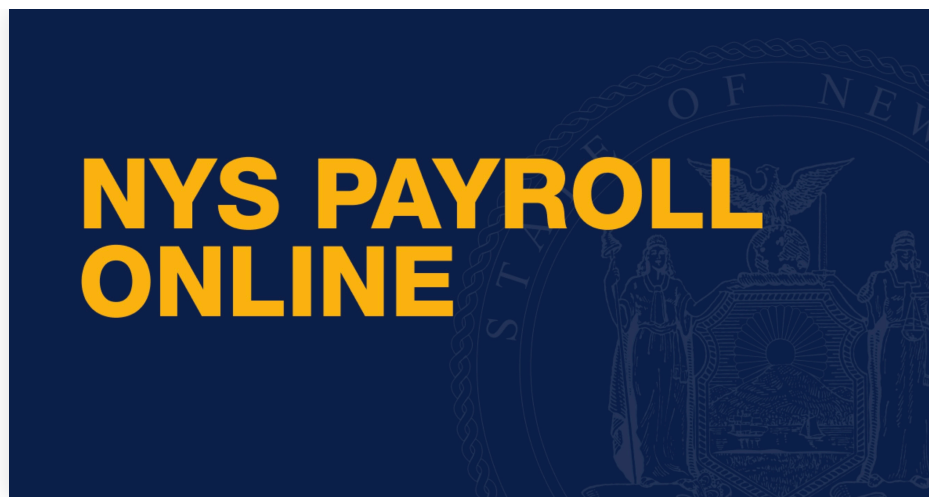
Create a plot of the **top 10 breeds** (by count) and comment on what you observe.

(a) Provide your `dplyr` + `ggplot2` code:

```
# YOUR CODE HERE
```

(b) Comment (2–3 sentences): Any surprises? What might explain the pattern?

Part 2. Data Transformation (`nyc_payroll_2025`)



For Questions 17–27, use `nyc_payroll_2025`.

For variable descriptions, see: Citywide Payroll Data (Fiscal Year) on NYC Open Data.

```
nyc_payroll_2025 <- read_csv("https://bcdanl.github.io/data/nyc_payroll_2025.zip")
```

Question 17. Base salary by borough (filter + summarise)

Compute the **mean** and **standard deviation** of `Base_Salary` for workers whose `Work_Location_Borough` is:

- "MANHATTAN"
- "QUEENS"

Report the two means and two SDs in your write-up.

```
# YOUR CODE HERE
```

Question 18. High base salary filter

Filter the data to keep only records where `Base_Salary` ≥ 100000 . Report how many rows remain.

```
# YOUR CODE HERE
```

Question 19. Distinct agency–title pairs

Select only **distinct** combinations of `Agency_Name` and `Title_Description`.

```
# YOUR CODE HERE
```

Question 20. Top paid by regular gross pay

Arrange employees by `Regular_Gross_Paid` in **descending** order and show the **top 10** rows with:

- `First_Name`, `Last_Name`
- `Agency_Name`
- `Title_Description`
- `Regular_Gross_Paid`

```
# YOUR CODE HERE
```

Question 21. Select + rename

Select `Title_Description` and rename it to `Title`. Also select `Agency_Name`, `First_Name`, `Last_Name`, and `Base_Salary`.

```
# YOUR CODE HERE
```

Question 22. Create new pay variables (`mutate()`)

Use `mutate()` to create two new variables:

- `Total_Pay = Regular_Gross_Paid + Total_OT_Paid + Total_Other_Pay`
- `OT_Share = Total_OT_Paid / Total_Pay`

Then show the first 10 rows of:

- `First_Name`, `Last_Name`
- `Agency_Name`
- `Base_Salary`
- `Total_Pay`, `OT_Share`

```
# YOUR CODE HERE
```

Question 23. Borough pay summary (`group_by()` + `summarise()`)

Using the variables you created in **Question 22**, group the data by `Work_Location_Borough` and compute:

- number of employees `n`
- mean `Base_Salary`
- mean `Total_Pay`
- mean `OT_Share` (ignore missing values)

Arrange the results by mean `Total_Pay` in **descending** order and show the summary table.

```
# YOUR CODE HERE
```

Question 24. Police Department overtime

Filter to `Agency_Name == "POLICE DEPARTMENT"` and arrange by `Total_OT_Paid`.

1. Show the **10 smallest** overtime values.
2. Show the **10 largest** overtime values.

```
# YOUR CODE HERE
```

Question 25. Per annum employees

Filter to `Pay_Basis == "per Annum"` and select only:

- `First_Name`, `Last_Name`, `Base_Salary`

```
# YOUR CODE HERE
```

Question 26. Borough then salary

Arrange by `Work_Location_Borough` (ascending) and then `Base_Salary` (descending). Then show the first 15 rows.

```
# YOUR CODE HERE
```

Question 27. Remove missing last names + count

Remove observations where `Last_Name` is missing (`NA`). Then report the remaining number of rows.

```
# YOUR CODE HERE
```

Part 3. Quarto Blogging (`ice_cream`)



Use the following dataset for your blog post.

```
ice_cream <- read_csv("https://bcdanl.github.io/data/ben-and-jerry-cleaned.csv")
```

Write and publish a blog post about **Ben & Jerry's ice cream** using `ice_cream`.

Your post must include:

- **At least 4** `ggplot2` figures
 - Each figure must be followed by a **short interpretation paragraph** (2–6 sentences).
- **At least 2** summary tables created with `group_by()` + `summarise()` (or `count()` + related verbs).
- Evidence of:
 - filtering,

- sorting,
- creating at least one new variable (`mutate()`), and
- using at least one **facet** (`facet_wrap()` or `facet_grid()`).
- A clear **data story** structure:
 1. a motivating question,
 2. what you did (briefly),
 3. what you found (with evidence),
 4. a takeaway / conclusion.

Important: Your plots and text should work together. Avoid disconnected charts.

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